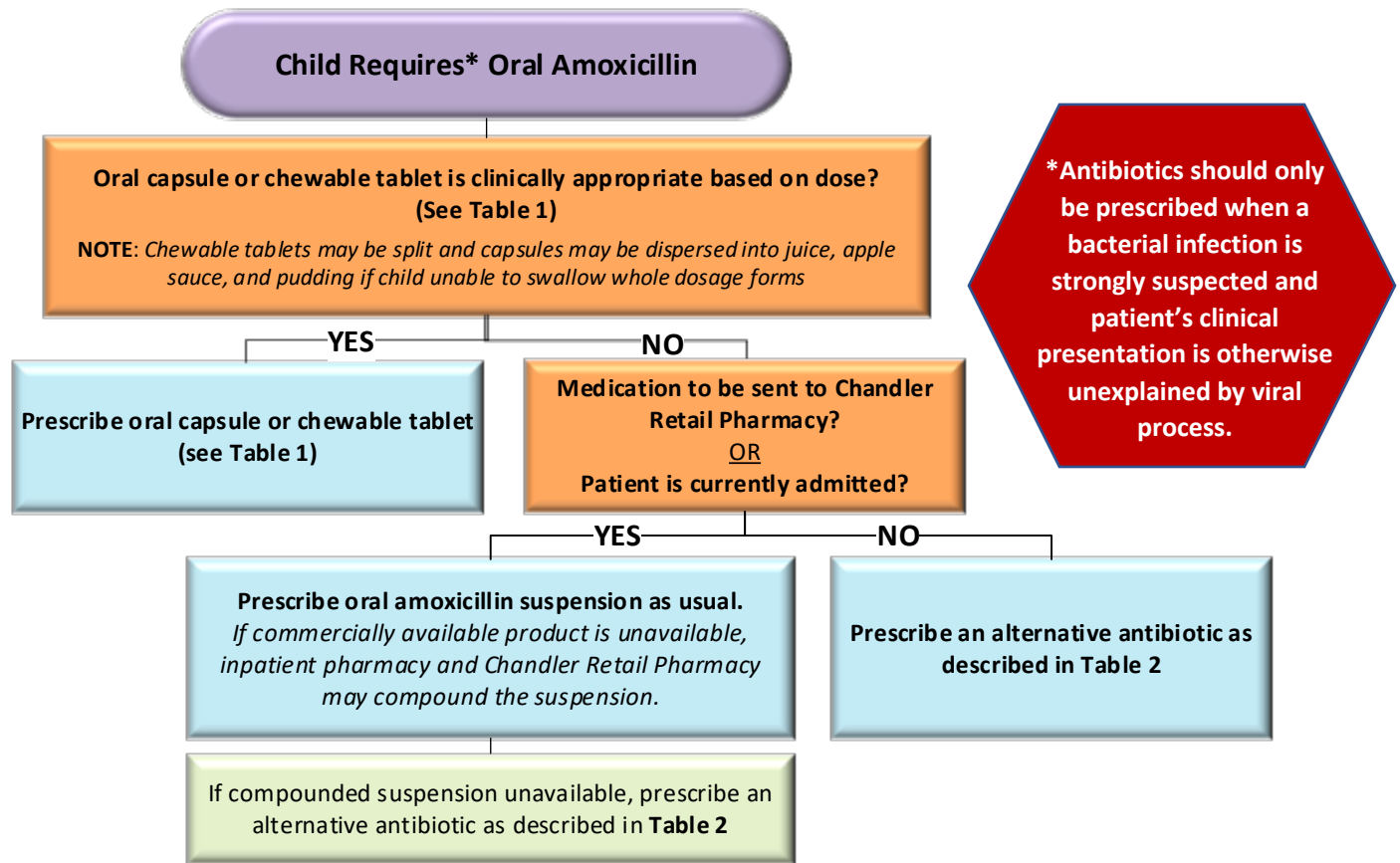


Antibiotic Interchange during Amoxicillin Suspension Shortage



The use of oral cephalosporins, in place of amoxicillin, for treatment of bacterial CAP, AOM, or rhinosinusitis is **strongly discouraged**. In addition to posing an increased risk for *C. difficile* infection and multi-drug resistance, second and third generation oral cephalosporins have poor oral absorption, decreased lung penetration, and provide inferior coverage of *S. pneumoniae*. See text below for more information.

Table 1: Amoxicillin Capsule and Tablet Interchange

Amoxicillin Oral Suspension Dose (Based on 15-45 mg/kg/dose)	125 mg Chewable Tablet* (Quantity of tablets to prescribe per dose)	250 mg Chewable Tablet* (Quantity of tablets to prescribe per dose)	250 mg Capsule^ (Quantity of capsules to prescribe per dose)	500 mg Capsule^ (Quantity of capsules to prescribe per dose)
50 mg to 75 mg	1/2 tablet	-	-	-
76 mg to 94 mg	-	-	-	-
95 mg to 149 mg	1 tablet	1/2 tablet	-	-
150 mg to 199 mg	1 and 1/2 tablets	-	-	-
200 mg to 299 mg	-	1 tablet	1 capsule	-
300 mg to 399 mg	-	1 and 1/2 tablets	-	-
400 mg to 574 mg	-	2 tablets	2 capsules	1 capsule
575 mg to 849 mg	-	3 tablets	3 capsules	-
850 mg to 1,100 mg	-	4 tablets	4 capsules	2 capsules

*Chewable tablets can be split

^Capsules may be opened and dispersed into juice, apple sauce, or pudding and immediately consumed

Table 2: Alternative Agent Interchange

Disease State	PREFERRED Alternative Agent
Acute Otitis Media	Amoxicillin/clavulanate* ≥1 month: 45 mg/kg/dose BID (max dose: 2,000 mg)
Pneumonia, Sinusitis, Respiratory Tract Infection	Amoxicillin/clavulanate* ≥1 month: 30 mg/kg/dose TID (max dose: 1,300 mg) <i>If ≥40 kg and able to swallow tablets, may use XR amoxicillin/clavulanate 2,000 mg BID (usually requires prior authorization)</i>
UTI TREATMENT	Cefadroxil ≥1 month: 15 mg/kg/dose BID (max dose: 1,000 mg)
UTI PROPHYLAXIS	Nitrofurantoin ≥4 weeks: 1-2 mg/kg/dose once daily (max dose: 100 mg) <i>In infants <4 weeks of age, attempt to secure amoxicillin suspension by contacting local pharmacies; if unable to obtain, contact Peds Antimicrobial Stewardship[‡] via EPIC secure chat for guidance</i>
Group A <i>Streptococcus</i> Pharyngitis	PO Penicillin VK <27 kg: 250 mg BID-TID ≥27 kg: 500 mg BID-TID <u>OR</u> IM Penicillin G Benzathine <27 kg: 600,000 units IM x 1 dose ≥27 kg: 1.2 million units IM x 1 dose
Other	Contact Peds Antimicrobial Stewardship[‡] via EPIC secure chat for considerations in uncommon circumstances

*Amoxicillin/clavulanate ES 600 mg/42.9 mg/5 mL is the preferred formulation. If difficulty obtaining supply of amoxicillin/clavulanate, clindamycin 10-13 mg/kg/dose TID (max dose: 450 mg TID) or doxycycline 2.2 mg/kg/dose BID (max dose: 100 mg BID) are considered appropriate alternatives for treatment of AOM, CAP, and respiratory tract infections. Of note, original data describing enamel staining are from studies with tetracycline. Recent comparative data in children suggest that doxycycline is NOT likely to cause visible teeth staining or enamel hypoplasia in children. The American Academy of Pediatrics recommend that doxycycline can be administered in any pediatric patient for durations ≤21 days without regard to the patient's age.⁽⁴⁾

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RATIONALE FOR AGENT SELECTION AND DOSING:

Oral amoxicillin is the drug of choice for treatment of community acquired pneumonia. Oral third generation cephalosporins are often inappropriately prescribed for treatment of CAP.⁵ The knowledge that IV ceftriaxone provides excellent coverage of *S. pneumoniae* often creates the misconception that all third generation cephalosporins provide coverage equivalent to amoxicillin. Rather, the contrary is true as oral third generation cephalosporins have poor oral absorption, decreased lung penetration, and provide inferior coverage of *S. pneumoniae* (greater than 60% of Serotype 19A pneumococci are resistant to oral cephalosporins).⁶⁻⁸ No oral cephalosporin provides activity against *S. pneumoniae* at the site of infection that equals high-dose amoxicillin.² Lastly, the pharmacokinetics of amoxicillin are best optimized when the total daily dose is divided at least three times daily. The recommendation to use twice daily dosing of amoxicillin is based on pharmacokinetic studies which documented the half-life of amoxicillin in middle ear fluid to be 4–6 hours, compared with a half-life of 1 hour in serum. Similar data supporting a recommendation for twice daily dosing of amoxicillin have not been collected for documented pneumococcal pneumonia in children.

References:

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